

JEFFREY SNOWDEN

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SUMMARY

Research scientist and analyst with eight years of experience designing surveys and measurement instruments, analyzing complex quantitative and qualitative data, and building dashboards and tools that make findings usable for nontechnical stakeholders. My experience includes R programming, psychometrics, multilevel modeling, research design, leading multi-site studies, and automating analysis pipelines.

PROFESSIONAL EXPERIENCE

BSCS Science Learning	2015 – Present
Research Scientist	Jan 2023 – Present
- Lead end-to-end survey and instrument development across large, multi-site studies, including design, validation, sampling, recruitment, data collection, analysis, and data-quality monitoring. - Analyze data using R and other statistical software, selecting from a broad range of statistical and psychometric methods, including multilevel models and Rasch measurement, to fit the needs of each study. - Developed a custom R package and function library that automate psychometric analysis pipelines and improve the efficiency, reproducibility, and consistency of analyses across projects. - Co-authored four peer-reviewed publications, including a 2024 article in <i>Science</i> and a book chapter on Rasch measurement, and presented research at AERA and NARST. - Develop interactive dashboards and tools that help nontechnical stakeholders explore complex datasets, understand findings, and use the results in project decisions. - Write the measurement and analysis sections of NSF grant proposals that have secured multi-year, multi-institution research funding. - Co-lead the organization-wide AI Task Force, conducting staff needs assessments and developing guidance for responsible AI use related to privacy, bias, and workforce impacts. - Co-chair the Institutional Review Board (IRB), overseeing research ethics, data privacy, and participant protection across the organization. - Lead research and evaluation teams on multi-institution studies, coordinating priorities, timelines, and budgets.	
Research Associate	Aug 2018 – Jan 2023
Cleaned, transformed, and modeled complex datasets while contributing to study design, instrument development, and quantitative and qualitative analyses across multiple research projects.	
Application Support Specialist	Dec 2016 – Aug 2018
Designed and maintained content and tooling for digital learning platforms, focused on usability and end-user needs.	

INDEPENDENT DATA PROJECTS

Latent Profile Explorer: Interactive application using synthetic employee-experience survey data to identify four engagement profiles and demonstrate model selection, validation, and nonresponse-bias analysis. Macroscopic Trends Viewer: Interactive time-series dashboard with brushable charts and custom markers Plumb Property-Tax Protest Tool: Searches Travis County appraisal data and generates a comps-based protest report.	
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EDUCATION

M.A., Research & Evaluation Methods , University of Colorado Denver	2020
<i>4.0 GPA; completed PhD-level quantitative methods coursework.</i>	
B.A., Philosophy & B.A., Communication , University of Colorado Colorado Springs	2016
<i>Magna Cum Laude, With Highest Distinction; 3.8 GPA.</i>	

SKILLS

Survey and measurement: instrument design and validation, scale construction, sampling, recruitment, Rasch measurement
Statistics and analysis: multilevel models, quasi-experimental design, latent profile analysis, mixed methods
Programming and tools: R, Stata, SPSS, Winsteps, HTML/JavaScript, Git/GitHub, LLM/RAG, prompt engineering
Data communication: dashboards, data visualization, technical reporting, communicating findings to nontechnical audiences
Leadership & governance: project and team leadership, research ethics and IRB, grant writing, budget coordination